Keerat Malhotra

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EDUCATION

Course	College / University	Year	CGPA / %
B.Tech (ITNS)	Netaji Subhas University of Technology	2028	8.34
CBSE Board (Class XII)	Guru Harkrishan Public School	2023	88.6%
CBSE Board (Class X)	BS International Sr. Sec. School	2021	89.2%

PROJECTS

- **CloudGuard – GCP Misconfiguration Scanner** GitHub| Live
  Tech Stack | Python, Google Cloud APIs (IAM, Cloud Storage, Compute Engine, Cloud Logging), Streamlit
  - Engineered a **GCP scanner** integrating the **Cloud Storage, IAM, Cloud Logging and Compute Engine APIs** across **4 independent detection modules**, normalizing results into a shared schema to compute a **severity-weighted 0–100 risk score**.
  - Architected **2 custom IAM roles** applying the **principle of least privilege** after identifying a permissions gap in GCP's predefined Viewer role (**storage.buckets.getIamPolicy** excluded), limiting the scanner's **service account** to **6 explicit read-only permissions.**
  - Developed a **protocol-aware auditor** for GCP **ingress firewall rules**, distinguishing **TCP/UDP/ICMP** traffic to detect **SSH, RDP and database ports** exposed to `0.0.0.0/0`, and eliminating false positives on **ICMP** traffic through explicit protocol-type filtering.
- **CampusBazaar – Anonymized Campus Marketplace** GitHub| Live
  Tech Stack | Next.js, Supabase (PostgreSQL, Auth, Realtime, Storage), Gemini API
  - Built a marketplace where **NSUT** students can trade notes, personal items and assignment help under **anonymous usernames**, with **Google OAuth 2.0** sign-in restricted to the **@nsut.ac.in** domain by a **database trigger**.
  - Implemented real-time, listing-scoped buyer-seller chat using **Supabase Realtime (WebSockets)**, securing every table with **PostgreSQL Row-Level Security (RLS)** so students can only read their own conversations and edit their own listings.
  - Integrated the **Gemini API** into a server-side **text-and-image moderation pipeline** via **API routes**, returning **structured JSON** verdicts on every new listing and routing flagged content to a **role-based admin dashboard** for **human review**.

TECHNICAL SKILLS

- **Programming Languages**: Java (Proficient), C++, Python
- **Coursework**: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems
- **Databases & Cloud**: GCP, PostgreSQL, MySQL
- **Tools & Integrations**: Git/GitHub, Docker, REST APIs, WebSockets

ACHIEVEMENTS

- **Grand Finalist, Smart India Hackathon 2025**(Link), built a real-time monitoring platform for NDMA disaster management trainings.
- **3rd Place, Xerocrypt CTF 2025**, among 150+ teams at NSUT.
- Solved **600+ DSA problems** across platforms like LeetCode (**1700+ rating**) and CodeChef (**2 star**).
- Achieved **98.64 percentile** in **JEE Main**.

CERTIFICATIONS

- **Google Cloud Cybersecurity Professional Certificate** – Coursera (Verify) **Oct 2026**

POSITIONS OF RESPONSIBILITY

- **DSA Head** | **DevComm, NSUT** **Jun 2026 – Present**
  Leading the DSA department, launching initiatives to upskill **50+ junior members** and reach **200+ students**.
- **Class Representative – B.Tech ITNS (Cohort of 72 Students)** **Dec 2024 – Present**
  Liaised between 72 students and faculty to coordinate timetables, academic schedules, and resolve queries.